Harshavardhan Raj Meerjumla

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2026	7.70
Omega Junior College	Class XII (TSBIE)	2020	67.00%
Bhashyam High School	Class X (TSBSE)	2017	92.00%

RELEVANT COURSES

Foundations of Machine Learning, Deep Learning, Visual Computing, Representation Learning, Linear Optimization, Advanced Computer Networks, Network Security, Advanced Data Structures and Algorithms, Fraud Analytics

POSITIONS OF RESPONSIBILITY

Teaching Assistant -- Algorithms, Data Structures at IIT Hyderabad

- Authored assignment questions, evaluated submissions, and mentored computer science undergraduate students.

Teaching Assistant -- Data Structures and Applications at IIT Hyderabad

- Helped non-CS students understand data structures through application-oriented doubt-solving sessions.

SKILLS

Languages/Database : Python, JavaScript, PHP, SQL

Machine Learning/Deep Learning : CNNs, Transformers, LSTMs, MLPs, Attention Mechanisms, scikit-learn, XGBoost, Random Forest

Tools/Frameworks : NumPy, PyTorch, scikit-learn, Pandas, Matplotlib, Databricks, Colab, Git, GitHub, Docker

Security/Networking : TCP/UDP Socket Programming, Wireshark, Burp Suite, tcpdump, Nmap, SQLmap

PROJECTS

Gossip-based Distributed Learning for Autonomous Driving [Github]

- Built a fully peer-to-peer federated continual-learning system for object detection on autonomous-driving imagery, enabling clients to exchange model weights without sharing raw data and achieving ~84% of centralized model mAP with negligible measured catastrophic forgetting.
- Designed and implemented the gossip protocol from scratch across three iterations using Faster R-CNN with a ResNet50-FPN backbone, incorporating loss-gated weight exchange and a custom replay buffer.
- Engineered an evaluation pipeline tracking mAP, mAR, forgetting, and BWT across diverse environmental conditions.
- Diagnosed a frozen RPN as the small-object recall bottleneck and improved centralized mAP from 0.257 to 0.285 through anchor retuning and RPN fine-tuning.

TagFed -- Reproduction & Extension [Github]

- Reproduced the CVPR 2024 paper *"Traceable Federated Continual Learning"*, achieving zero measured catastrophic forgetting while reducing per-round communication by 230× through logit-only distillation.
- Traced a GPU memory leak consuming ~4 GB of VRAM per run to a retained computation graph and modernized 25+ incompatible PyTorch 1.x calls for PyTorch 2.x.
- Removed the paper's test-time task-ID assumption by designing a task-agnostic inference method using max-softmax confidence across task heads, with no observed loss in accuracy.

TransactiWar - Secure P2P Banking Web App [Github]

- Built a full-stack PHP/MySQL banking application with a five-member team for a live red-team/blue-team cybersecurity exercise; placed 2nd overall with zero successful attacks against our application during the offensive round.
- Implemented secure authentication using bcrypt password hashing, database-enforced uniqueness constraints, and parameterized queries to prevent SQL injection across authentication endpoints.
- Hardened session management against fixation and hijacking by regenerating session IDs on login, enforcing HttpOnly and SameSite=Strict cookies, and implementing configurable idle/absolute timeouts.
- Implemented brute-force protection using rate-limited lockouts and integrated CSRF token validation across all authentication endpoints.

Fraud Detection with Adversarial Anomaly Models (AnoGAN & BiGAN) [Github]

- Built an unsupervised fraud detection pipeline in PyTorch on 284K+ credit card transactions (0.17% fraud), training AnoGAN and BiGAN on normal-only data to handle extreme class imbalance
- Implemented latent-space optimization and joint encoder-discriminator training, scoring anomalies via reconstruction error and discriminator feature-matching
- Built a leak-free evaluation pipeline with validation-tuned thresholds and a rank-normalized model ensemble, benchmarked on ROC-AUC, precision/recall, and F1

EXTRACURRICULAR

Gaming: Valorant, Minecraft, Stellaris

Musical Instruments: Guitar, Piano

Rubik's Cube: <20-second solves